

BYTE'S CODE CITY — FULL CHEAT SHEET

Method in Motion · Byte the Data Tracker

BYTE'S METHOD

IDENTIFY THE TYPE → FIND THE PATTERN → ORGANIZE IT CLEARLY

LESSON 1 — DATA BASICS

DATA

Any piece of information a computer stores or works with.

DATA TYPE

The category a value belongs to: string, number, or boolean.

STRING

Text wrapped in quotes. Example: "Byte".

NUMBER

A numeric value without quotes. Example: 100.

Remember: "100" is a string; 100 is a number.

BOOLEAN

Only true or false. No quotes.

VARIABLE

A labeled container that stores data.

Example: let score = 100;

LABEL → CONTAINER → VALUE

In let score = 100:

Label = score

Container = the variable created with let

Value = 100

Value type = number

PREDICT → CHANGE → RUN

Predict the output first. Change a value. Identify its type. Predict again. Run to check.

LESSON 2 — PATTERNS & SEQUENCES

ARRAY

An ordered list of values.

Example: ["pink", "cyan", "gold"]

SEQUENCE

Values arranged in a specific order.

INDEX

The position of an item in an array.

ZERO INDEXING

Arrays count from 0.

First item = index 0

Second = index 1

Third = index 2

RETRIEVE BY INDEX

colors[0] returns the first item.

colors[2] returns the third item.

PATTERN

A sequence that repeats or follows a rule.

Example: 2, 4, 6, 8 follows "add 2."

BYTE'S ARRAY PROCESS

Check the order → Find the rule → Use the index to retrieve the clue.

LESSON 3 — ORGANIZING INFORMATION

OBJECT

Organized data stored as key-value pairs.

Example: { name: "Byte", level: 4 }

KEY

The label for a piece of information.

VALUE

The information connected to a key.

KEY-VALUE PAIR

A label connected to its data. Example: level: 4.

PROPERTY

One key-value entry inside an object.

DOT NOTATION

Retrieves a property.

profile.name returns "Byte".

profile.level returns 4.

LOOSE VS. ORGANIZED DATA

Loose: "Byte", "Data Tracker", 4

Organized: { name: "Byte", role: "Data Tracker", level: 4 }

SORT

Arrange data into an order.

GROUP

Collect similar items together.

LESSON 4 — BYTE'S DATA CHALLENGE

ARRAY OF OBJECTS

A list where every item is its own organized profile.

MESSY DATA FILE

"Zen", 3, true, "Byte", 4, false

STEP 1 — IDENTIFY THE TYPE

"Zen" = string

3 = number

true = boolean

"Byte" = string

4 = number

false = boolean

STEP 2 — FIND THE PATTERN

name → level → ready status, repeated.

STEP 3 — ORGANIZE IT CLEARLY

```
[  
{ name: "Zen", level: 3, ready: true },  
{ name: "Byte", level: 4, ready: false }  
]
```

COMBINE INDEX + DOT NOTATION

crew[1].name

crew = array

[1] = second object

.name = name property

Result = "Byte"

CAPSTONE RECALL

Can I identify strings, numbers, and booleans?

Can I explain a variable?

Can I remember arrays start at 0?

Can I retrieve by index?

Can I spot a sequence or pattern?

Can I identify keys and values?

Can I retrieve a property with dot notation?

Can I organize repeated data into an array of objects?

Can I use Byte's method from start to finish?

SYMBOLS REFERENCE

[] — Array brackets; create an array or retrieve by index.

{ } — Object braces; hold key-value pairs.

: — Connects a key to a value.

. — Dot notation; retrieves an object property.

" " or ' ' — Quotes; wrap strings.

= — Assigns a value in the lesson examples.

QUICK RECALL

"Byte" → string

42 → number

false → boolean

let score = 100 → variable storing a number

["A","B","C"][0] → "A"

profile.level → retrieves level

crew[1].name → second object's name

BYTE'S FINAL RULE

IDENTIFY THE TYPE → FIND THE PATTERN → ORGANIZE IT CLEARLY